The following B.E.S.T Standards will be covered in this unit:

MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

- *Clarification 1: Within this benchmark, the expectation is to perform all operations with integers.*
- *Clarification 2: Refer to Properties of Operations, Equality and Inequality (Appendix D).*

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

- *Clarification 1: Properties include associative, commutative and distributive.*
- *Clarification 2: Refer to Properties of Operations, Equality, and Inequality (Appendix D).*

MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.

- *Clarification 1: Problems include the variable in multiple terms or on either side of the equal sign or inequality symbol.*

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

- *Clarification 1: Instruction includes linear expressions in the form $ax \pm b$ or $b \pm ax$, where a and b are rational numbers.*
- *Clarification 2: Refer to Properties of Operations, Equality, and Inequality (Appendix D).*

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

- *Clarification 1: Instruction includes using properties of operations accurately and efficiently.*
- *Clarification 2: Instruction includes linear expressions in any form with rational coefficients.*
- *Clarification 3: Refer to Properties of Operations, Equality, and Inequality (Appendix D).*

Unit 13, Lesson 1: Evaluating Expressions

Benchmark

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

Learning Targets

- ☐ I can apply order of operations to an algebraic expression.
- ☐ I can evaluate expressions that include exponents.

13.1.1 Warm-Up: Evaluating Numeric Expressions

This math expression is dividing the internet.

What is the value of $8 \div 2(2+2)$?

1. Do you agree with Alex or Ms. B? Explain or show why using order of operations.

2. Find the value of the expression. Show your work.

$$4 - 2 \times 6 \div 3 + 7$$

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations. Grouping symbols such as parentheses are used to indicate which operations should be performed first. The following order is used.

Step 1: Perform operations within grouping symbols.
Step 2: Evaluate exponents.
Step 3: Multiply and divide in order from left to right $(\rightarrow)$.
Step 4: Add and subtract in order from left to right $(\rightarrow)$.

1. Three students completed the warm-up. The steps in each student's process are shown.

Leila's Work	***Fabian's Work***	***Jaylah's Work***
$4-2\times 6\div 3+7$	$4-2\times 6\div 3+7$	$4-2\times 6\div 3+7$
$2\times 6\div 3+7$	$4-12\div 3+7$	$4-12\div 3+7$
$12\div 3+7$	$4-4+7$	$4-4+7$
$4+7$	$4-11$	$0+7$
11	-7	7

a. Complete the statements about which student correctly applied the given steps of the order of operations.

○ Leila
○ Fabian
○ Jaylah

evaluated the numerical expression correctly.

She first performed all
○ multiplication and division
○ addition and subtraction
in order from left to right and then performed all
○ multiplication and division
○ addition and subtraction
in order from left to right.

b. With your partner, discuss the work of each student, Leila, Fabian, and Jaylah. Two of these students made a mistake somewhere. Circle the mistakes that you and your partner found.

2. Omari evaluated Expression A and Expression B. His work is shown.

Expression A	***Expression B***
$12+24\div 6-2\times 2$	$12+24\div(6-2\times 2)$
$12+4-2\times 2$	$12+24\div(6-4)$
$12+4-4$	$12+24\div(2)$
$16-4$	$12+12$
12	24

a. With your partner, discuss the two original expressions that Omari evaluated and his corresponding answers.

b. Omari's partner gave him a 👍 on both of his answers. Why did Omari arrive at two different values when evaluating these expressions?

Like numeric expressions, algebraic expressions can be evaluated using order of operations when given a specific value for the ***variable***(s).

In an algebraic expression, a ***variable*** is a letter that represents a number. It is called a **vari**able because the value used to replace the letter can ***vary***.

3. When evaluating algebraic expressions, first substitute the given _________ for the variable(s). Then, apply the given steps for order of operations.

In mathematics, **evaluate** is an instruction that indicates the answer is a **value**.

4. Evaluate $\frac{n}{6} + 2$ when $n = 12$.

In numeric expressions, multiplication can be represented in different ways. Each expression below is a way to represent "three times five."

3×5 $\qquad$ $3 \cdot 5$ $\qquad$ $3(5)$

In algebraic expressions, the multiplication symbol, $\times$, could be confused with the variable x. Therefore, writing a number next to a variable is a better way to note multiplication when using variables.

$3x$ means "three times x."

Multiplication of a number and a variable could also be represented using the multiplication dot, $3 \cdot x$, or parentheses, $3(x)$, but $3x$ will be the most common way.

5. Explain why $3 \cdot 5$ cannot be written as 35.

6. What is the value of $3x$ if $x = 5$?

When substituting values into algebraic expressions, it is helpful to put the substituted value in **parentheses** when the variable is being multiplied by a known value.

7. Find the value of $7m - (3 + b - 12) \div 2$ if $m = 4$ and $b = 15$. Show your work.

13.1.3 Collaboration: Order of Operations

An order of operations was first noted in First Year Algebra by Webster Wells and Walter Wilson Hart. The textbook was published in 1912.

1. The problem below is an example from *First Year Algebra.*

Example. Find the numerical value of the expression,

$$4\,ab + \frac{6\,c}{b} - d^3,$$

when $a = 4, \ b = 3, \ c = 5, \ d = 2.$

a. What is the numerical value of the expression? Show your work.

b. Check your work with your partner.

c. The work shown in the textbook is below.

$$4\,ab + \frac{6\,c}{b} - d^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2 \cdot 2 \cdot 2$$
$$= 48 + 10 - 8 = 50.$$

What is the difference between your work and the work from the book?

An order of operations became more important when programmers needed a set of rules to program a computer to find values of numeric expressions. In 1883, Ada Lovelace described in her book about Charles Babbage's analytical engine, the first computer, that codes could be created so the device could handle letters and symbols along with numbers.

13.1.4 Guided Instruction: Evaluating Expressions With Exponents

The work in the book, *First Year Algebra,* showed 2^3 written out as multiplication. This method can get complex as shown in the table.

$(-3)^3 + 2^6 \div -16 + 5$
Exponents written as multiplication
$((-3)(-3)(-3)) + (2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) \div -16 + 5$

$(-3)^3 + 2^6 \div -16 + 5$
Evaluate expressions with exponents first
$-27 + 64 \div -16 + 5$

- When writing $(-3)^3$ and 2^6 using multiplication, both were written inside a set of parentheses.
- In order of operations, parentheses indicate that the operations inside should be done ___________ other steps.
- Therefore, evaluating exponents is done before other multiplication and division when following the order of operations.

1. Evaluate $(-3)^3 + 2^6 \div -16 + 5$.

2. Evaluate each expression, then determine if the expressions are equivalent.

Expression A	Expression B	Equivalent?
$(-3)^4$	$-(3^4)$	
$-(9)^2$	$(-9)^2$	
(-5^2)	$(-5)^2$	
$26 - 4^3$	$26 - (-4)^3$	
$17 + (-2^4)$	$17 - (2^4)$	

3. Describe a strategy that can be used when evaluating expressions with exponents that include a negative sign or a subtraction sign.

4. Elizabeth evaluated $b + a^3 \cdot 7 - 9$ when $a = -4$ and $b = 17$. Her work is below.

   a. Circle Elizabeth's error.

$$17 + (-4)^3 \cdot 7 - 9$$
$$17 + (-64) \cdot 7 - 9$$
$$(-47) \cdot 7 - 9$$
$$-329 - 9$$
$$-338$$

   b. Which step of the order of operations did she not apply correctly?

   Ⓐ Evaluate expressions with exponents.
   Ⓑ Perform operations inside grouping symbols.
   Ⓒ Perform multiplication and division in order from left to right.
   Ⓓ Perform addition and subtraction in order from left to right.

The order of performing computations is to first work within grouping symbols using the ***order of operations***. Then, simplify terms with exponents. Next, while reading from left to right, perform multiplication and division in the order in which it appears. Finally, while reading from left to right, perform addition and subtraction in the order in which it appears.

13.1.5 Try It: Discovering the Importance of Left to Right

1. Mia is evaluating the expression, $x^3 \div r \cdot (k^2 - 5)$ for $k = 3, r = -8$ and $x = 4$. Her first few steps are shown.

$$4^3 \div -8 \cdot (3^2 - 5)$$
$$64 \div -8 \cdot (9 - 5)$$
$$64 \div -8 \cdot 4$$

   a. Mia thinks that she could finish her work by either finding $64 \div -8$ or $-8 \cdot 4$ first. Show the work for both possibilities to determine if Mia is correct.

   b. Was Mia correct when she stated that both methods would work? Use the steps of the order of operations to justify whether or not Mia is correct.

2. Answer the following problems.

a. Evaluate $r \cdot W \div x \div m$ when $m = 2$, $r = -4$, $W = -60$, and $x = -10$. Show your work.

b. Evaluate $(r \cdot W) \div (x \div m)$ when $m = 2$, $r = -4$, $W = -60$, and $x = -10$. Show your work.

c. How did the use of parentheses change the answers?

13.1.6 Wrap-Up: Error Analysis

1. Nadine and Nora evaluated the expression $r^3 \div x \cdot c$ for $c = 9, r = -6$ and $x = -3$. Both students made errors in their work shown. Circle their errors.

Nadine's Work	Nora's Work
$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-216 \div (-3) \cdot 9$ $-216 \div -27$ 8	$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-18 \div (-3) \cdot 9$ $6 \cdot 9$ 54

2. Write a short explanation to each student to explain the mistake and what she needs to remember about evaluating expressions in the future.

To Nadine:	*To Nora:*

Unit 13, Lesson 2: Evaluating Algebraic Expressions Using Substitution

Benchmark

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

Learning Target

- ☐ I can evaluate algebraic expressions using substitution.

13.2.1 Warm-Up: Order Explorer

1. Place a digit in each box to create an expression with the largest possible value. Use any of the digits 1 to 5, once each at most.

$$\square^{\square} + \square \times \square$$

2. What is the value of your expression?

3. On a scale of 1 to 10, how confident are you that your expression is the largest possible value?

4. Explain your strategy for choosing and placing the numbers as you did.

13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers

1. Brae evaluated the expression $7b + 12 - (5-3)^2$ for $b = -2$. Her work is shown.

   a. Evaluate her work and determine where she made mistakes.

$$
\begin{gathered}
7b + 12 - (5-3)^2 \\
7 - 2 + 12 - (5-3)^2 \\
7 - 2 + 12 - (-2)^2 \\
7 - 2 + 12 + 4 \\
5 + 12 + 4 \\
17 + 4 \\
21
\end{gathered}
$$

   b. Help Brae understand and learn from her mistakes. Write a brief note to Brae explaining what her mistakes were and how she could learn from the mistakes.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

   c. Read the note your partner wrote and offer suggestions on ways to improve their note.

   d. Using the word bank below, complete the three reminders that could help Brae.

inside	outside	parentheses

- When substituting values into algebraic expressions, it is helpful to put the substituted value(s) in ________________.
- When evaluating expressions with exponents, if the negative is ____________ the parentheses, the negative indicates subtraction, or the opposite of the value found by raising the value inside to a power.
- If the negative is ________________ the parentheses, it indicates that the number is a negative number being raised to a power.

   e. Verify your answers with your partner and, if necessary, resolve any differences.

2. Complete the following problems.

   a. Evaluate the expression $-8m - 4q^2 + \frac{s}{3}$ where $m = 4$, $q = -3$, and $s = -24$.

   b. Evaluate the expression $3s - 4q - m^2$ where $m = 4$, $q = -3$, and $s = -24$.

   c. Evaluate the expression $-m^3 - 3q + (s - q)$ where $m = 4$, $q = -3$, and $s = -24$.

   d. All of the expressions should have the same result. Check your work with your partner.

13.2.3 Your Turn!

1. What is the value of $5(6x + 7) - 3y^2$ if $x = -1$ and $y = 5$?

2. Evaluate the expression $3d^4 - \frac{W}{3}$ where $d = 4$ and $W = -75$.

3. Find the value of $-2(b + 4) + 3a - (7 - a)^5$ if $a = -3$ and $b = 5$.

4. Evaluate the expression $c^3 - 5k + (h - k)$ where $c = -9$, $h = 29$, and $k = -17$.

5. Evaluate each expression.

a. $8 \div p - 16 \div (-5 + 7)$ where $p = 4$

b. $13 + \frac{6}{y} - y^3$ if $y = -3$

c. $4g - 12 + (h - 7) \cdot 3^2$ where $g = -7$ and $h = 6$

d. $w \div t \cdot H \div y^2$ where $H = -3$, $t = 7$, $w = 84$, and $y = -2$

13.2.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding.

I'm so confident, I can explain evaluating algebraic expressions using substitution to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute a value

I tried hard, but I am finding this challenging. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute in a value

Unit 13, Lesson 3: Applying Properties of Operations

Benchmark

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

Learning Target

- [ ] I can apply the associative, commutative, and distributive properties to generate equivalent expressions.

13.3.1 Warm-Up: Mental Math

Ulrich finds the product of $5 \cdot 107$ in his head. Ulrich wrote the following explanation of his thinking:

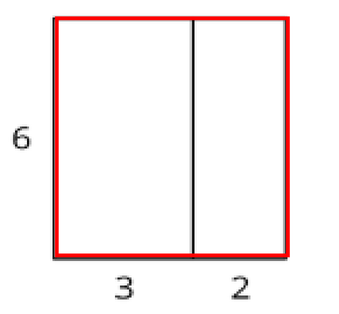
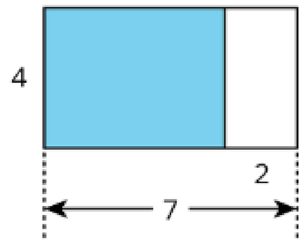
> Because 107 is equal to $100 + 7$, I can write $5 \cdot 107$ as $5(100 + 7)$. I know that $5 \cdot 100$ is 500 and that $5 \cdot 7$ is 35. Then, I add 500 and 35 to get 535.

1. Use Ulrich's reasoning to write each in the form $a(b + c)$. Then, use that form to find the product.

Multiplication	$5 \cdot 104$	$5 \cdot 97$	$5 \cdot 130$
$a(b + c)$			
Product			

13.3.2 Exploration: Representing Area of a Rectangle

1. Work with your partner to answer the questions.

Rectangle A	Rectangle B
	

a. Rectangle A is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of Rectangle A.

- ☐ $6 \cdot 5$
- ☐ $6 \cdot 3 \cdot 2$
- ☐ $6(3 + 2)$
- ☐ $6 \cdot 3 + 2$
- ☐ $6 + 3 + 2$
- ☐ $6 \cdot 3 + 6 \cdot 2$

b. With your partner, discuss which expressions are in the form $a(b + c)$.

c. Rectangle B is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of the blue rectangle inside Rectangle B.

- ☐ $4 \cdot 5$
- ☐ $4 \cdot 7 \cdot 2$
- ☐ $4(7 - 2)$
- ☐ $4 \cdot 7 - 4 \cdot 2$
- ☐ $4 \cdot 7 + 4 \cdot 2$
- ☐ $4 \cdot 2 - 4 \cdot 7$

d. With your partner, discuss which expressions are in the form $a(b + c)$.

2. Complete the following problems.

a. Circle all of the expressions that have the same value as $-4(x+7)$ if $x = {}^{-}13$.

$-4x+7$	$(x+7)\cdot-4$	$-4x+-4\cdot 7$	$4(x-7)$	${}^{-}28-4x$

b. With your partner, discuss whether or not the expressions you circled would change if the x-value were a different number. Summarize your decision and reason.

13.3.3 Guided Instruction: Applying Properties of Operations

The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property "distributes" a factor to all terms within a set of parentheses.

1. Complete the table. The first row has been completed.

$\boldsymbol{a(b+c)}$	$\boldsymbol{ab+ac}$	**Value**
${}^{-}8(50+9)$	${}^{-}8(50)+{}^{-}8(9)$	${}^{-}472$
	$17(20)+17(3)$	
$(100-7)\cdot{}^{-}6$		
$(40+72)29$		
	${}^{-}11(48)-{}^{-}11(96)$	

2. In two of the rows the expression had a subtraction sign instead of an addition sign. Discuss with your partner how $a(b-c)$ can be rewritten using addition. Write a summary of your discussion.

3. Use the expressions below to show that the distributive property of multiplication over addition works with rational numbers.

$a(b + c)$	$ab + ac$	**Equivalent?**
$\frac{1}{2}(^{-}14 + 50)$	$\frac{1}{2}(^{-}14) + \frac{1}{2}(50)$	
$1.2(12 + 50)$	$1.2 \cdot 12 + 1.2 \cdot 50$	

Mathematicians determine a property exists when expressions are equivalent, no matter what number is used. A property can then be used with variables.

Expressions are **equivalent** if they have the same value, no matter what number is substituted in for each of the variables.

4. Write an equivalent expression for $8(x + 2)$.

5. Write an equivalent expression for $\frac{1}{2}(x - 16)$.

The **commutative property** states that the order in which an operation is performed does not change the value of the expression.

6. Use the pairs of expressions below to determine the operations for which the commutative property is true.

Operation	**Example Expressions**		**Equivalent?**
Addition	$\frac{5}{12} + \frac{3}{4}$	$\frac{3}{4} + \frac{5}{12}$	
Subtraction	$\frac{3}{4} - \frac{11}{2}$	$\frac{11}{2} - \frac{3}{4}$	
Multiplication	$\frac{5}{6} \cdot \frac{2}{3}$	$\frac{2}{3}\left(\frac{5}{6}\right)$	
Division	$\frac{7}{12} \div \frac{1}{4}$	$\frac{1}{4} \div \frac{7}{12}$	

The commutative property only applies to

_________________ and __________________________.

7. Use the commutative property of multiplication to write an expression equivalent to $5 \cdot \frac{3}{20} \cdot 4\frac{4}{5}$.

8. Rewrite $x + 9$ using the commutative property.

9. Rewrite $(^{-}7)(x)$ using the commutative property.

Commutative Property of Addition

$a + b = b + a$

Commutative Property of Multiplication

$a \times b = b \times a$

The **associative property** states that the way in which numbers are grouped does not change the value of the expression.

10. Use the pairs of expressions below to determine the operations for which the associative property is true.

Operation	Example Expressions		Equivalent?
Addition	$\left(\frac{8}{5} + \frac{3}{10}\right) + \frac{17}{10}$	$\frac{8}{5} + \left(\frac{3}{10} + \frac{17}{10}\right)$	
Subtraction	$\frac{7}{2} - \left(\frac{3}{4} - \frac{1}{4}\right)$	$\left(\frac{7}{2} - \frac{3}{4}\right) - \frac{1}{4}$	
Multiplication	$\frac{2}{3} \cdot \left(\frac{3}{5} \cdot \frac{1}{9}\right)$	$\left(\frac{2}{3} \cdot \frac{3}{5}\right) \cdot \frac{1}{9}$	
Division	$\left(\frac{1}{5} \div \frac{1}{10}\right) \div \frac{1}{2}$	$\frac{1}{5} \div \left(\frac{1}{10} \div \frac{1}{2}\right)$	

The associative property applies only to

_____________________ and _________________________.

11. Rewrite $(3.25 + 9.8) + 7.2$ using the associative property.

12. Rewrite $(x + 15) + {}^{-}11$ using the associative property.

13. Rewrite $({}^{-}2)(8 \cdot x)$ using the associative property.

14. Rewrite $6\big((x + 3) + 2\big)$ using the associative property.

Associative Property of Addition

$(a + b) + c = a + (b + c)$

Associative Property of Multiplication

$(a \times b) \times c = a \times (b \times c)$

13.3.4 Wrap-Up: Ticket Out the Door

1. Write an equivalent expression that uses each given property to complete the table.

Expression	Property	Equivalent Expression
$-2\big(4 + (x + {}^{-}7)\big)$	Commutative Property	
$-2\big(4 + (x + {}^{-}7)\big)$	Associative Property	
$-2(4 + x + {}^{-}7)$	Distributive Property	

Unit 13, Lesson 4: Generating Equivalent Algebraic Expressions

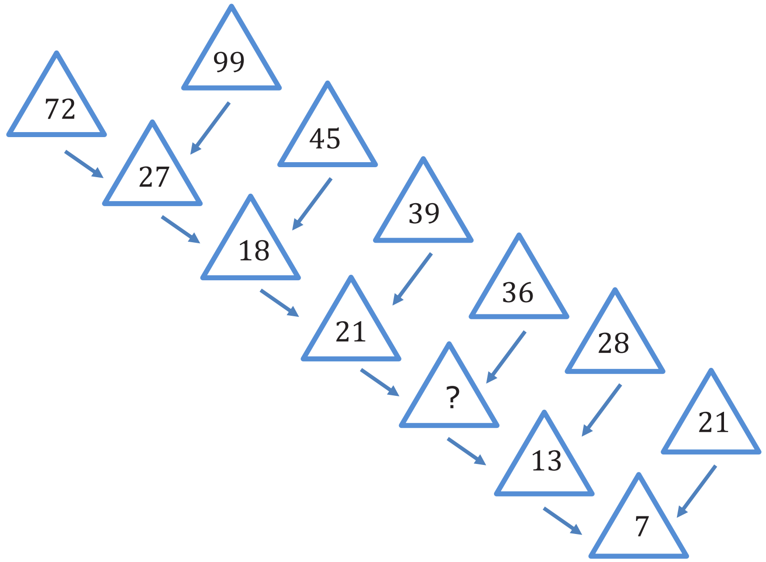
Benchmark

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

Learning Target

☐ I can apply properties of operations to generate equivalent expressions.

13.4.1 Warm-Up: Missing Number



1. What is the first number you tried for the missing number?

2. Did your first number work for the rest of the pattern?

3. What strategy did you try if your first number did not work?

13.4.2 Guided Instruction: Generating Equivalent Algebraic Expressions

1. Consider the following expressions.

$8 \cdot x + 8 \cdot 9$ $\qquad$ $8(x + 9)$ $\qquad$ $8x + 9$

   a. Which of these expressions are equivalent?

   b. Which property of operation justifies that the expressions are equivalent?

2. Write two equivalent expressions. Name the property or properties you used to create each equivalent expression.

3. For each expression, use the associative and commutative properties to generate an equivalent expression.

Original Expression	Associative Property	Commutative Property
$(3x + 5) + 4z$		
$8 \cdot (-1 \cdot 13d)$		

4. Rewrite $-11b + 3(10k + 1)$ using the distributive property.

5. Rewrite $-11b + 3(10k + 1)$ using the commutative property of multiplication.

6. Rewrite $-11b + 3(10k + 1)$ using the commutative property of addition.

7. Makayla and Soledad both rewrote $(-5h+9)\cdot -14$ using the commutative property of multiplication. Verify that their expressions are equivalent by substituting a value in for h. Show your work in the table.

Makayla	**Soledad**
$-14(-5h+9)$	$(h\cdot -5+9)\cdot -14$
$h=$ ______	$h=$ ______

13.4.3 Try It: Writing Equivalent Algebraic Expressions

1. Select the expressions that are equivalent to $(2x+7)+(5y-3)$.

- ☐ $(5y-3)+(7+2x)$
- ☐ $(3-5y)+(7+2x)$
- ☐ $7+(5y+2x)-3$
- ☐ $(7+5y)+(2x-3)$
- ☐ $5y+(7+2x)-3$

2. Write two equivalent algebraic expressions for each expression below.

a. $(4+9)\cdot x$

b. $3t\cdot 4$

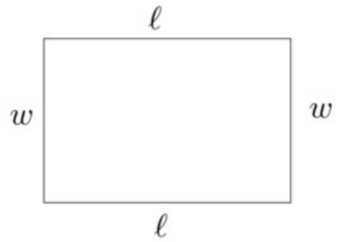
c. $-7x+3-14$

d. $3(^{-}9p-4)+7$

13.4.4 Your Turn!

1. The students in Ms. Justino's class are writing expressions for the perimeter of a rectangle of side length, ℓ, and width, w. The expressions for four students are in the table.

Student	Expression
De'Andre	$2(\ell + w)$
Summer	$\ell + w + \ell + w$
Abm	$2\ell + w$
Luzarie	$2\ell + 2w$



Ms. Justino told the group that one of the expressions is not equivalent. Write a short explanation to the student to show why their expression is not equivalent to the other expressions.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

2. Match equivalent expressions by drawing an arrow. If an expression does not have a match, write an equivalent expression to match it in one of the blank boxes provided.

Column A
$-8(x+3)$
$-8x \cdot 3$
$(8x \cdot -3) \cdot 2$
$(3 + {}^{-}8x) + 2$
$2 + {}^{-}8x$

Column B
$2 + ({}^{-}8x + 3)$
$3 \cdot -8x$
$-24 - 8x$

13.4.5 Wrap-Up: Error Analysis

Tangela was asked to circle all of the expressions equivalent to $6(y+1)$.

Her work is shown below.

$6y+1$	$6y+7$	$6(y)+1(y)$	$6(y)+6(1)$	$6y+6$

1. Determine if Tangela correctly identified all of the equivalent expressions. If she is missing an expression, circle the expression(s) she is missing.

2. If Tangela made an error, then explain her error and list the equivalent expressions.

Unit 13, Lesson 5: Writing Algebraic Expressions

Benchmark

MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

Learning Targets

- ☐ I can translate a written description into an algebraic expression.
- ☐ I can translate a real-world description into an algebraic expression.

13.5.1 Warm-Up: Bell Work

1. Two diagrams are shown. One represents $3 + 5 = 8$. The other represents $5 \cdot 3 = 15$.

   Complete the table by writing the equation that matches each diagram in the space provided and labeling the length of each diagram.

Equation		
Diagram	$\underbrace{\boxed{3}\boxed{3}\boxed{3}\boxed{3}\boxed{3}}_{\boxed{}}$	$\underbrace{\boxed{3}\boxed{\quad 5 \quad}}_{\boxed{}}$

2. Draw a diagram that represents each equation.

   a. $4 + 3 = 7$

   b. $4 \cdot 3 = 12$

13.5.2 Guided Instruction: Making Sense with Models

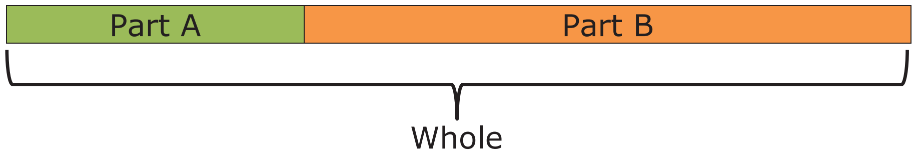
An ***algebraic expression*** is a mathematical statement containing numbers, operators, and at least one unknown, represented by *a* variable.

1. Complete the statement.

   An expression ○ does ○ does not have an equal sign or inequality symbol.

Bar models help connect the action of a problem to an algebraic expression.

Part-Part-Whole Model



The part-part-whole model can be used to show the following actions.

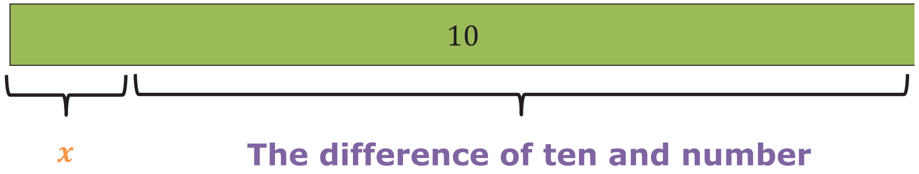
Action	Equation
Combining or joining	Part A + Part B = Whole
Taking away or separating	Whole − Part A = Part B Whole − Part B = Part A

2. "The sum of a number and three" can be shown as the length of the whole bar where one part is n and the other part is 3.

The sum of a number and three

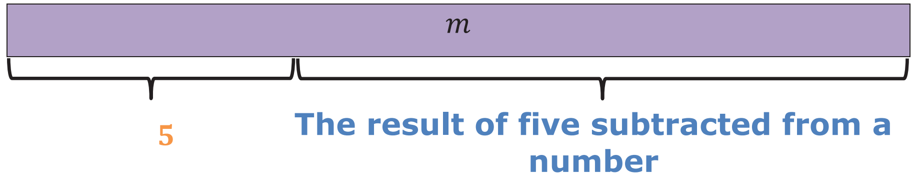
Write an algebraic expression for "The sum of a number and three."

3. "The difference of ten and a number" can be shown where the whole bar is 10 and one part is a number represented by a variable, x. The difference is the other part.



Write an algebraic expression for "The difference of ten and a number."

4. "Five subtracted from a number" is shown with the whole bar as a number represented by a variable, m, and one part is 5. The other part is the result of 5 being subtracted from the number, m.

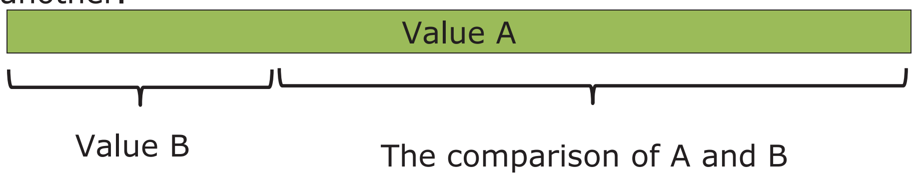


Complete the statement.

The algebraic expression for "Five subtracted from a number, m" is ________________.

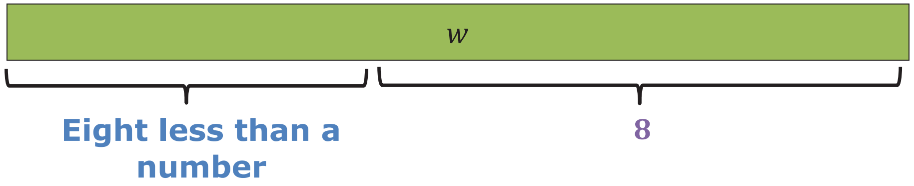
Addition/Subtraction Comparison Model

Comparison models are much like part-part-whole models. They show how much more or less one value is than another.



How much **more** Value A is **than** Value B, or
How much **less** Value B is **than** Value A

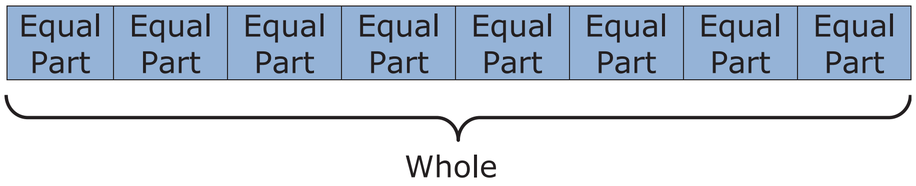
5. Write an algebraic expression for the bar model shown.



Algebraic expression: ____________________

Equal Parts – Whole Model

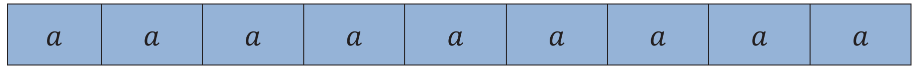
The Equal Parts – Whole Model shows the whole sectioned into equal parts.



The equal parts-whole model can be used to show the following actions.

Action	Equation
Repeated addition (multiplication)	Equal part + … + equal part = whole Equal part × number of parts = whole
Equal sharing	Whole ÷ number of parts = equal part
Partitioning	Whole ÷ equal part = number of parts

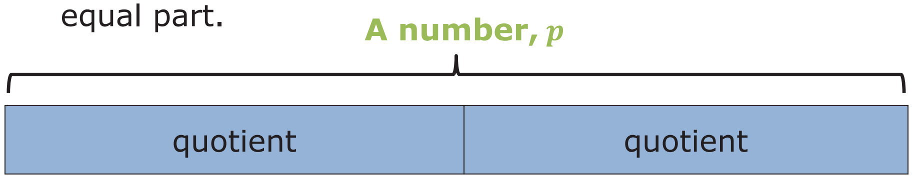
6. "The product of nine and a number" can be shown as 9 equal parts with size of "a number" represented with the variable, a.



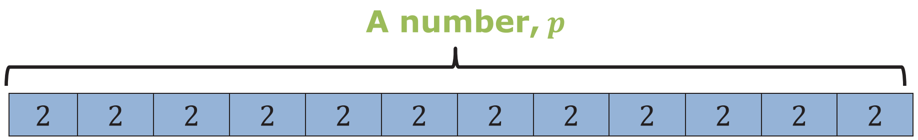
The product of nine and a number

Write an algebraic expression for "the product of nine and a number."

7. "The quotient of a number and two" could be shown as "a number" represented by the variable, p, sectioned off into two equal parts. The quotient is the size of each equal part.



"The quotient of a number and two" could also be shown as "a number" represented by the variable p, sectioned off into equal parts with size of 2 units. The quotient is the number of equal parts.



What is the algebraic expression for "The quotient of a number and 2?"

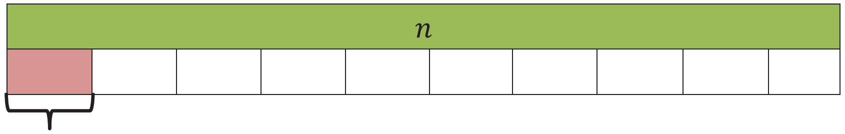
13.5.3 Collaboration: Matching Activity

Part 1

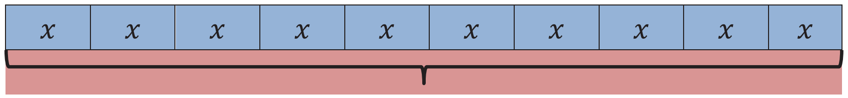
Match each written description to the appropriate bar model, where the red portion is the result of the written description. Write the letter of the bar model beside the corresponding word description.

Written Description	Bar Model
1. Ten subtracted from a number	
2. A number divided by ten	
3. The sum of a number and ten	
4. A number times ten	

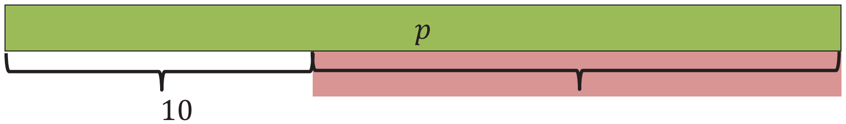
A.



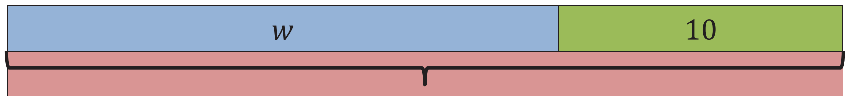
B.



C.



D.



Part 2

Turn and talk with your partner to compare your matches. Then, work with your partner to write an algebraic expression that represents the written description and bar model.

Written Description	Algebraic Expression
1. Ten subtracted from a number	
2. A number divided by ten	
3. The sum of a number and ten	
4. A number times ten	

13.5.4 Guided Instruction: Writing Algebraic Expressions From Descriptions of Real-World Contexts

Algebraic expressions can be used to represent real-world situations.

1. Misty has 34 pairs of shoes arranged on a number of shelves. The same number of shoes are on each shelf. She has h number of shelves.

   a. Write an algebraic expression that represents this situation.

   b. What does the algebraic expression represent?

2. Martin buys 8 pounds of apples at a price per pound to make pie. Each pound costs p dollars.

   a. Write an algebraic expression to represent this situation.

   b. What does the algebraic expression represent?

3. Paige is 6 inches taller than her brother. Paige's brother is m inches tall.

   a. Write an algebraic expression to represent this situation.

   b. What does the algebraic expression represent?

13.5.5 Try It: Writing Algebraic Expressions

Write an algebraic expression to represent each situation.

1. Joe is 9 years younger than Drew, who is n years old.

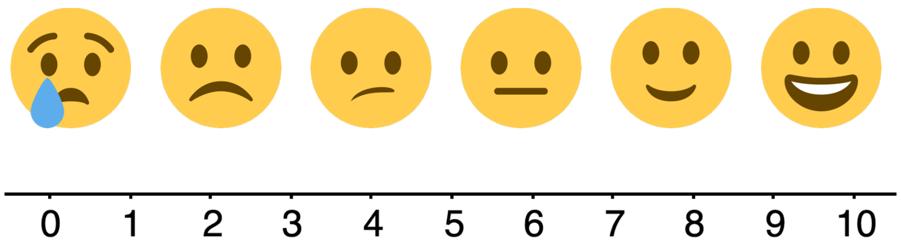
2. Jenna's salary, s, is lowered by $\$120$.

3. Maria has 12 cupcakes and wants to split them among f friends.

4. Target has 8 cookie packages with c cookies in each.

13.5.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of translating a written description into an algebraic expression.



2. Complete the statement.

I chose this emoji because ______________________________

____

____

Unit 13, Lesson 6: Writing Descriptions of Real-World Contexts From Algebraic Expressions

Benchmark

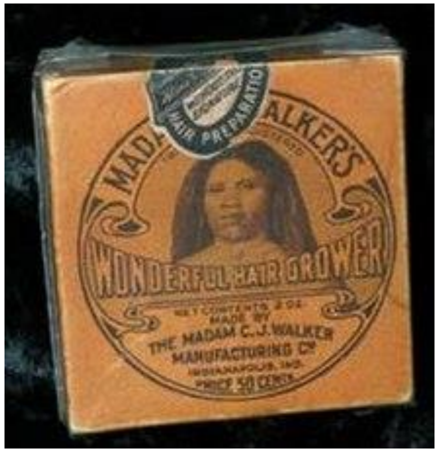
MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

Learning Target

- ☐ I can create a real-world context from an algebraic expression.

13.6.1 Warm-Up: Self-Made Millionaire

Madam C.J. Walker was the first African American self-made millionaire. She made her fortune by inventing and selling specialized hair products, including an innovative Edge Control product that sold for $\$0.35$ each beginning in 1905.



1. Create an algebraic expression to describe the total revenue Madam C.J. Walker earned for selling any number of Edge Control products, e.

13.6.2 Activity: Matching Real-World Contexts to Expressions

1. Match each real-world situation with the correct expression.

A. Ryan has 20 songs. His friend bought him x more songs. How many songs does Ryan have?

B. Karen has x flowers. Helen has 20 times more flowers than Karen. How many flowers does Helen have?

C. George earned 20 points in a card game. Bruno earned x fewer points than George. How many points did Bruno earn?

D. Maria has 20 books and wants to put an equal number of books on x bookshelves. How many books go on each shelf?

1. $20x$
2. $x + 20$
3. $\frac{20}{x}$
4. $20 - x$

13.6.3 Collaboration: Writing Real-World Contexts

1. For each expression, create a real-world scenario that the algebraic expression could represent.

Expression	Your Real-World Scenario	What does the variable represent?
$r - 10$		
$21y$		
$\frac{100}{a}$		
$17 + s$		
$57 - r$		
$48w$		

2. With your partner, pick four expressions from Question 1 and use them to complete the following table.

Expression	Your Partner's Scenario	Does the Expression Accurately Represent the Scenario?	Similarities Between Your and Your Partner's Scenarios for the Expression
		☐Yes ☐No	
		☐Yes ☐No	
		☐Yes ☐No	
		☐Yes ☐No	

13.6.4 Activity: Gallery Walk

Several scenarios are displayed. For each scenario, write an algebraic expression to match.

Scenario	Expression

13.6.5 Wrap-Up: Ticket Out the Door

1. Frank has a plant that measured h inches tall last week. This week the plant is 8 inches tall.

   a. Circle the expression that represents the number of inches, h, the plant grew in a week.

   $8 - h$ $h - 8$

   b. Explain your choice.

   c. For the expression not chosen, describe a real-world situation that the expression might represent.

Unit 13, Lesson 7: Adding and Subtracting Linear Expressions With Integer Coefficients

Benchmark

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

Learning Target

☐ I can use algebra tiles to add and subtract linear expressions.

13.7.1 Warm-Up: Career in Robotics

Tessa Lau is the Chief Robot Whisperer for a robotics company. The company creates robot helpers for the service industry. Some of their robots perform tasks like delivering items to guests at a hotel. Tessa has studied applied physics, computer science, artificial intelligence, and human-computer interaction. She loves hands-on experiments, building things, and the speed of creating with computers. Through programming and engineering, one of her primary jobs is to look at a robot in context and make sure the robots behave well in human environments.

1. Have you ever considered a career as an engineer?

2. How do you think a robot could be helpful to you in your everyday life?

3. Tessa's advice is "Don't listen to someone who tells you that you can't do something." Describe a time someone told you that you were not able to do something.

13.7.2 Refresher: Adding and Subtracting Integers With Algebra Tiles

Algebra tiles, like integer chips, can be used to model adding and subtracting integers. Each small square represents a positive 1 (blue) or a negative 1 (red).

Integer Chips	Algebra Tiles
+ −	1 -1

Adding a positive 1 (blue) and a negative 1 (red) represents a zero pair because $1 + {}^{-}1 = 0$, as shown.

$\boxed{1}\boxed{-1} = 0$

1. Describe why this diagram represents the value 0.

1	-1
1	-1
1	-1

A ***zero pair*** is a pair of numbers whose sum is zero.

2. Complete the following problems.

a. Complete the table.

Numeric Expression	Algebra Tiles Representation	Value
$-7 + 2$		
$6 - 8$		
$-1 - 5$		
$4 - (^{-}3)$		

b. For some of the representations with algebra tiles, zero pairs had to be added to the representation. Put an asterisk (*) next to the expressions where this was the case.

c. Explain why zero pairs had to be added to the representation in these cases.

13.7.3 Guided Instruction: Algebra Vocabulary

There are some fundamental definitions that are necessary to know for working with algebraic expressions.

A ***term*** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The expression $0.5 - 4y$ has two terms.

0.5	$-4y$
term 1	term 2

One of the terms in this expression is a constant term.

1. Which term is the constant term? Explain.

An ***expression*** is a mathematical statement containing numerals, operators, grouping symbols, and symbols or variables for unknown values. An expression does not contain an equal sign or inequality symbol.

For example, $4 + 5$, $3x - 2$, and $6x - 7y + 11$ are all expressions.

A ***coefficient*** is the number or constant that multiplies a variable in an algebraic expression. If no number is specified, the coefficient is 1.

In the expression $4xy$, 4 is the coefficient.

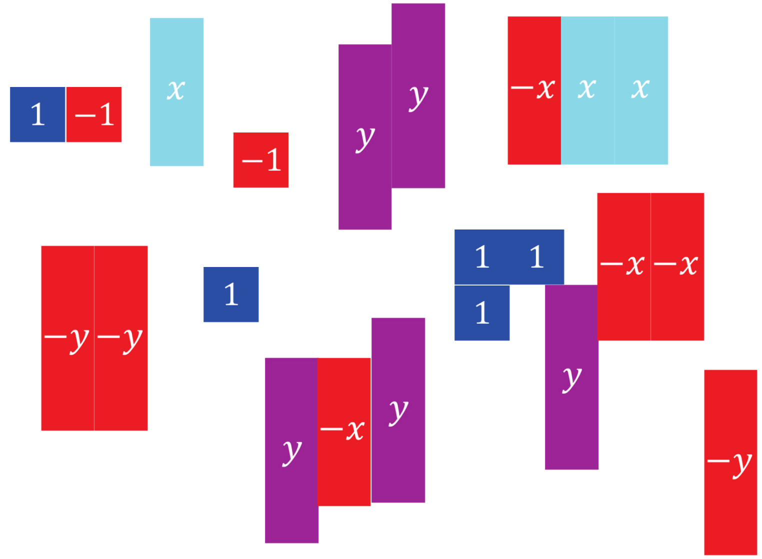
In the expression x, 1 is the coefficient.

In the expression $-8y$, -8 is the coefficient.

Variable terms have coefficients while constant terms do not.

13.7.4 Exploration: What Are Like Terms?

1. Here is a collection of algebra tiles.



   a. Complete the table below.

Type of Algebra Tile	Number of Tiles
x	
$(-x)$	
y	
$(-y)$	
1	
(-1)	

b. Use the definitions of coefficient and term to complete the expression by writing the coefficient or constant term that matches the number of tiles of each type.

$\_\_\_\_x + (-\_\_\_\_x) + \_\_\_y + (-\_\_\_\_y) + \_\_\_\_ + (-\_\_\_\_)$

c. Draw the number of algebra tiles of each type under the term in the expression.

d. Circle the zero pairs in your drawing.

e. Complete the expression below after eliminating the zero pairs.

$$\_\_\_\_x + \_\_\_y + \_\_\_\_$$

Like terms are terms with the exact same variable(s) raised to the same exponent.

A ***constant*** is a fixed numerical value with no variables. All constants are like terms.

2. Answer the following questions about like terms.

a. Classify the following terms.

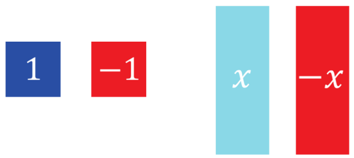
	Like Terms	**Unlike Terms**
y and 6	○	○
$3x$ and 3	○	○
$5x$ and $-3x$	○	○
$10, -2, 7.5$	○	○

b. How did you determine whether the terms were like or unlike?

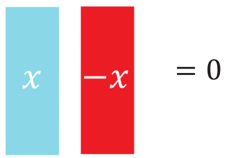
c. Compare answers with your partner.

13.7.5 Guided Instruction: Adding and Subtracting Expressions With Algebra Tiles

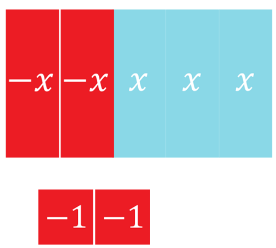
The algebra tiles shown represent x and $-x$ as rectangles, and integers as squares. Red represents the negative of that variable or constant, while the blue shades represent the positive.



Variables also form zero pairs, as shown.



1. Complete the table by crossing out the zero pairs and recording how many there are. Write the resulting algebraic expression after eliminating the zero pairs.

Algebra Tiles	Number of Zero Pairs	Algebraic Expression
Representation: 		

2. Draw a representation of the following expressions with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

Algebraic Expression and Representation	Equivalent Algebraic Expression
$x - 3x + 1$ **Representation:**	
$-4 + 3y - y + 2$ **Representation:**	

13.7.6 Guided Practice: Adding and Subtracting Expressions With Algebra Tiles

1. Consider the expressions $2x$ and $x + 2$.

   a. Draw a model of $2x$ and $x + 2$ with algebra tiles.

Expression	Algebra Tiles Model
$2x$	
$x + 2$	

   b. How are they different?

   c. Are $2x$ and $x + 2$ equivalent algebraic expressions?

13.7.7 Your Turn!

Write an equivalent expression by combining like terms.

1. $2x + 4x =$ ____________________

2. $-y + 7 + 4y - 1 =$ ____________________

3. $x - x + x - x + 1 =$ ____________________

4. $8x - 4 - 4x =$ ____________________

5. $2x - 4x =$ ____________________

13.7.8 Wrap-Up: Cool Down

1. For each key word, check the box in the column that describes your knowledge of the word. Then answer the questions or complete the statements in the last column.

Key Word	I already knew the meaning of this word.	I've heard this word before, but I was not sure what it meant.	Questions
Expression	☐	☐	An example of a numeric expression is ______________. An example of an algebraic expression is ______________.
Term	☐	☐	In the expression $8x - 9 + 2y$, how many terms are there? ______
Coefficient	☐	☐	In the expression $8x - 9 + 2y$, what are the coefficients? ________________
Equivalent	☐	☐	When working with algebraic expressions, the word "equivalent" means ________________ ________________ ________________.

Unit 13, Lesson 8: Adding Linear Expressions With Rational Coefficients

Benchmark

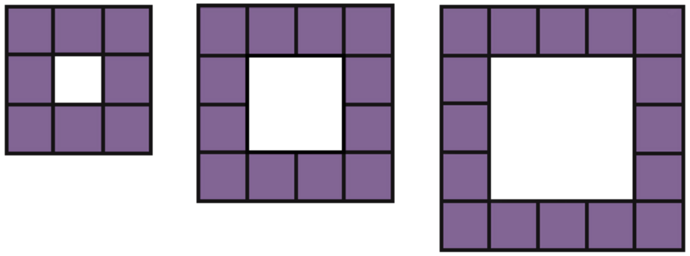
MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

Learning Target

☐ I can add linear expressions with rational coefficients.

13.8.1 Warm-Up: Visual Patterns

Three figures are shown.



1. What pattern do you see?

2. Explain how many small purple squares you think will be in the next figure.

13.8.2 Exploration: Which One Doesn't Belong?

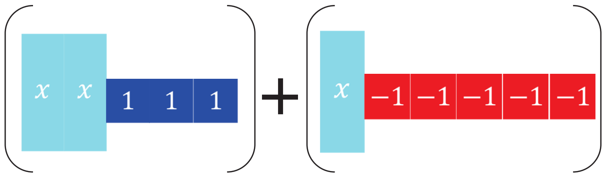
1. Which expression does not belong with the other three? Explain your reasoning.

Expression A	Expression B
$7 - 5x - 3$	$\frac{3}{2}x + 6 - 2 - 6\frac{1}{2}x$
Expression C	**Expression D**
$-5x + 7.25 - 4.25$	$-2x - 3x + 4$

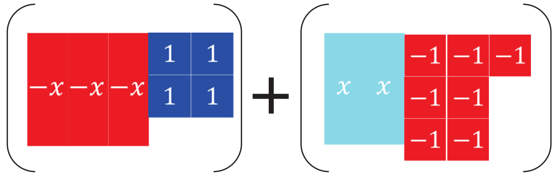
13.8.3 Collaboration: Adding Expressions With Algebra Tiles

Algebra tile representations of an expression are given. Work with your partner to find the sum by following the steps below.

1.

On the lines, write the algebraic expression for each representation.	 __________ + __________
Rearrange the tiles to group like tiles. Draw the representation.	
Eliminate any zero pairs to find the sum. Write the sum.	

2.

On the lines, write the algebraic expression for each representation.	 __________ + __________
Rearrange the tiles to group like tiles. Draw the representation.	
Eliminate any zero pairs to find the sum. Write the sum.	

13.8.4 Guided Instruction: Adding Linear Expressions

When adding two expressions, rearrange the terms so that like terms are grouped together. Then add the coefficients of the like variable terms together and add the constants. Zero pairs are eliminated in this process.

Subtracting an integer is the same as adding the opposite. For example, the expression $9 - 13$ can be rewritten as $9 + -13$.

1. Find the sum of the two expressions.

a.

Original Expression	$(2x + 6) + (5x + 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	____x + ____x + ____ + ____
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	____x + ____

b.

Original Expression	$(-2x + 6) + (5x - 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	____x + ____x + ____ + ____
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	____x + ____

13.8.5 Collaboration: Adding Linear Expressions

1. Work with your partner to find the sum of the two expressions in simplest form.

a.

Original Expression	$4 + (8 - 0.4x)$
Regrouped Expression	
Simplified Expression	

b.

Original Expression	$\left(3\frac{1}{2}x - \frac{3}{5}\right) + \left(2\frac{1}{4}x + \frac{1}{3}\right)$
Regrouped Expression	
Simplified Expression	

c.

Original Expression	$(-5y + 2) + (5y - 8)$
Regrouped Expression	
Simplified Expression	

d.

Original Expression	$(0.7 - 2.4x) + (4.1x - 5.32)$
Regrouped Expression	
Simplified Expression	

e.

Original Expression	$\left(\frac{3}{4}w - 5.5\right) + \left(-\frac{1}{2}w + 4\right)$
Regrouped Expression	
Simplified Expression	

13.8.6 Your Turn!

1. Find the sum of each expression using the fewest terms possible.

   a. $(x + 7) + (x - 10)$

   b. $(3x - 5) + (5 + 6x)$

   c. $\left(1\frac{6}{7} - y\right) + \left(4y + \frac{1}{7}\right)$

   d. $(4w - 6) + (-6w - 1)$

   e. $(9 - 4.6x) + (-5.3x - 7)$

   f. $\left(-\frac{5}{2}k - 7\right) + \left(\frac{1}{5}k + 4\right)$

13.8.7 Wrap-Up: Error Analysis

1. Your friend finds the sum of $\left(-2\frac{1}{2}x + 7.1\right) + (4x - 5.2)$ to be $-\frac{3}{2}x + 1.9$.

   Write a detailed note to your friend explaining why their answer is correct or why it is incorrect.

Unit 13, Lesson 9: Subtracting Linear Expressions With Rational Coefficients

Benchmark

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

Learning Targets

- ☐ I can subtract linear expressions with rational coefficients.
- ☐ I can apply properties of operations to add and subtract linear expressions with rational coefficients.

13.9.1 Warm-Up: Bell Work

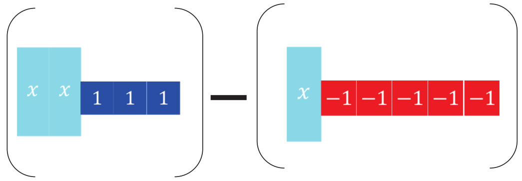
Combine like terms.

1. $(3y + 4.6) + (-5y - 3.25)$

2. $\left(6\frac{3}{4}x - 7\right) + \left(-\frac{5}{2}x + 7\right)$

13.9.2 Exploration: Subtracting Expressions With Algebra Tiles

1. Algebra tile representations are given below. Use the algebra tiles to find the difference.

On the lines, write the algebraic expression for each representation.	 __________ − __________
Rearrange the tiles to show subtraction of like tiles together. Draw the representation.	
If necessary, create any zero pairs needed for subtraction.	
Find the difference by subtracting (taking away) like tiles. Write the algebraic representation of the difference.	

13.9.3 Collaboration: Subtracting Expressions

1. Noah, Lin, Jada, and Andre are rewriting the expression $(x + 8) - (4 - 9x)$ as an equivalent expression with fewer terms. Each student described their work.

   Discuss with your partner which students made an error. Describe each student's error and correct the error.

   a. Noah says, "I dropped the parentheses, combined like terms, and ended up with $-8x + 4$."

   $$(x + 8) - (4 - 9x)$$
   $$x + 8 - 4 - 9x$$
   $$-8x + 4$$

   b. Lin says, "I simplified inside the parentheses and ended up with $13x$."

   $$(x + 8) - (4 - 9x)$$
   $$(8x) - (-5x)$$
   $$8x + 5x$$
   $$13x$$

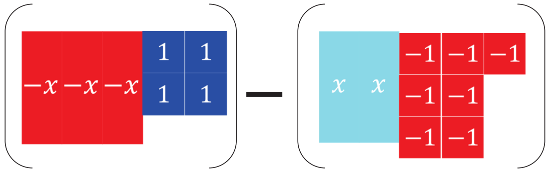
   c. Jada says, "I know that subtracting is adding the opposite, and ended up with $4 + 10x$."

   $$(x + 8) - (4 - 9x)$$
   $$(x + 8) + (-4 + 9x)$$
   $$x + 8 - 4 + 9x$$
   $$4 + 10x$$

   d. Andre says, "I used the distributive property since there is an invisible (-1) in front of the second parentheses to distribute, but I ended up with $10x + 4$."

   $$(x + 8) - (4 - 9x)$$
   $$(x + 8) - 1(4 - 9x)$$
   $$x + 8 - 4 + 9x$$
   $$10x + 4$$

2. When subtracting expressions, use the distributive property to add the opposite. With the algebra tile representations below, model finding the difference by adding the opposite.

On the lines, write the algebraic expression for each representation.	 __________ − __________
Rewrite subtracting to adding the opposite by changing (or flipping) the sign of the 2nd expression. Draw the representation.	
Combine any zero pairs to find the sum. Write the sum.	

13.9.4 Guided Instruction: Subtracting Linear Expressions

1. To subtract linear expressions,

   - add the ________________, or additive inverse of each term in the expression.

   OR

   - distribute a ___________ to each term in the expression.

2. Complete the table to show both methods.

Additive Inverse Method	**Applying the Distributive Property**
$(4x + 5) - (-2x + 6)$	$(4x + 5) - (-2x + 6)$

3. When adding and subtracting, parentheses are used to group expressions. Why is the use of parentheses especially important when subtracting?

13.9.5 Guided Practice: Subtracting Linear Expressions With Rational Coefficients

1. Find the difference of each expression.

a. $(3y + 3.5) - (8.2y - 6)$	b. $\left(m - \frac{3}{8}\right) - \left(3\frac{1}{4} - m\right)$
c. $\left(-2 + \frac{9}{2}y\right) - \left(5.3y - 5\frac{1}{2}\right)$	d. $(-7.5t - 3) - \left(7 + 3\frac{1}{8}t\right)$

2. What is the result when $-3x + 4.48$ is subtracted from $-2.2x + 3$?

   a. Write the algebraic expression.

   b. Rewrite the expression without parentheses.

   c. Simplify the expression by rewriting it using the fewest terms possible.

13.9.6 Collaboration: Adding and Subtracting Linear Expressions

1. Two students made errors when rewriting the expression $(6x + 9) - (7 - x) + (8x + 11)$.

Zoe's Work	Patricia's Work
$(6x + 9) - (7 - x) + (8x + 11)$	$(6x + 9) - (7 - x) + (8x + 11)$
$(6x + 9) - (7x + 18)$	$(5x + 2) + (8x + 11)$
$-x - 9$	$13x + 13$

   a. Working with your partner, write an explanation to Zoe that describes her error.

   b. Write an explanation to Patricia that describes her error.

   c. Simplify the expression $(6x + 9) - (7 - x) + (8x + 11)$.

13.9.7 Your Turn!

1. Rewrite each expression by combining like terms.

   a. $(1 - 4y) - (2.5y + 7) - (-0.3y + 3)$

   b. $\left(3\frac{1}{2}x - 4\right) + (x - 3.2) - \left(-\frac{1}{4}x + 9\right)$

   c. $-(2r + 5) - (6 - 3r) + (10r - 4)$

13.9.8 Wrap-Up: Ticket Out the Door

Write the following expressions using the fewest possible terms.

1. $\left(5x - 13\frac{1}{4}\right) - \left(4\frac{1}{2} - 5x\right)$

2. $(0.7 - 2.4x) + \left(-5.32 + 4\frac{1}{10}x\right)$

Unit 13, Lesson 10: Equivalent Linear Expressions

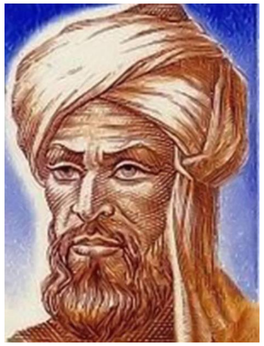
Benchmark

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

Learning Target

☐ I can verify using substitution that linear expressions are equivalent.

13.10.1 Warm-Up: Origins of Algebra



The word **algebra** comes from the Arabic word الجبر, or al-jabr, which means, "the restoring of broken parts." The word algebra comes from the title of a book written by a Persian mathematician named Muhammad ibn Musa al-Khwarizmi in the 9^{th} century.

1. Based on what you know about algebraic expressions and equations, explain why you think this meaning of the Arabic word makes sense to describe algebra.

13.10.2 Refresher: Evaluating Algebraic Expressions

Evaluate each expression for the given value of x.

1. $-3x - 8$ for $x = 5$

2. $4 - 2(x^2 - 7)$ for $x = -1$

3. $(4.5 + x) \cdot 3$ for $x = -\frac{5}{2}$

13.10.3 Collaboration: Solving Equivalent Expressions Using Substitution

1. With your partner, complete the table by choosing who will be Partner A or Partner B to determine if the expressions are equivalent.

Partner and Value	$8x + 4x - 2$	$2(6x - 1)$	Equivalent?
A $x = 2$			○ yes ○ no
B $x = 5$			○ yes ○ no
A $x = -5$			○ yes ○ no
B $x = (-2)$			○ yes ○ no

2. Complete the statement to describe the results in the last column of the table.

   Depending on the value being substituted into the expressions, the evaluated expressions
   - ○ always
   - ○ sometimes
   - ○ never

   had the same value as each other.

3. What do the results tell you about the expressions $8x + 4x - 2$ and $2(6x - 1)$?

4. Rewrite $8x + 4x - 2$ by combining the like terms.

5. Use the distributive property to rewrite $2(6x - 1)$.

6. Discuss with your partner how the two expressions verify the results from your table.

13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution

The two expressions $8x + 4x - 2$ and $2(6x - 1)$ are equivalent, but not equal. The expressions $12x - 2$ and $12x - 2$ are equal.

1. What is the difference between **equivalence** and **equality**?

Algebraic expressions are **equivalent** if they have different terms but are equal when evaluated using the same value for the variables. Algebraic expressions are **equal** if they have the same terms and operations.

2. Determine whether $5 - 3(2x + 1)$ and $4x + 2$ are equivalent using substitution of numerical values.

a. Complete the table.

x	$5 - 3(2x + 1)$	$4x + 2$
What is the value when $x = 0$?		
What is the value when $x = \frac{1}{2}$?		
Choose another value for x to substitute.		

b. Explain if the expressions are equivalent.

c. Explain why the expressions were equal when "0" was substituted for x.

d. Discuss with your partner the advantages and disadvantages of using "0" to substitute for a variable to check for equivalence.

3. Determine whether $3 + x(4 - 2)$ and $x(3-7) + 5$ are equivalent using substitution of numerical values.

   a. Complete the table.

x	$3 + x(4 - 2)$	$x(3-7)+5$
What is the value when $x = 1$?		
What is the value when $x = 0$?		
Choose another value for x to substitute.		

   b. How many values are best to use when substituting values in for the variable to check for equivalency?

13.10.5 Guided Practice: Equivalent Expressions Using Substitution

1. Consider the three expressions below. Determine whether the expressions are equivalent by substituting the given value of x.

   a. Complete the table.

x	$4x - \frac{1}{2}(6+8x)$	$8x-3$	$-(4x+3)+4x$
$x = 3$			
$x = -2.75$			
$x =$ _____			

b. What conclusion can you draw about the expressions from the information in the table?

2. Determine whether the expressions $\frac{2}{5}(15x - 30y) + 10x$ and $4(4x - 2y)$ are equivalent.

x and y	$\frac{2}{5}(15x - 30y) + 10x$	$4(4x - 2y)$
$x = -2$ $y = 0$		
$x =$ _____ $y =$ _____		

13.10.6 Your Turn!

1. Determine which expressions are equivalent by substituting the same values for the variable.

A $(5x - 3) - \frac{3}{4}(8 + 2x)$	B $-5\frac{1}{4}(4x - 1)$
C $9 - 6\left(\frac{-2}{3}x + 4\right)$	D $\frac{3}{2}x - 9 + 2x$
E $\left(\frac{30}{2}\right)(-1) + 4x$	F $6 - 12\frac{1}{2}x - \frac{3}{4} - 8.5x$

2. Complete each statement using the expressions from the table.

__________________ is equivalent to __________________

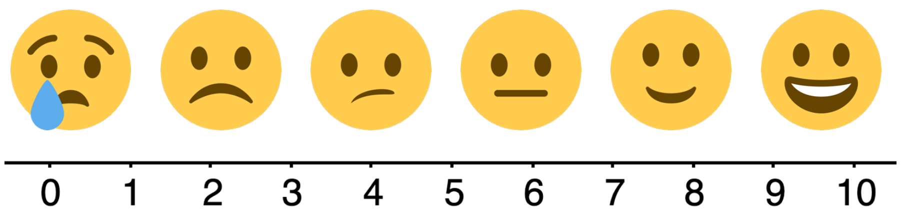
__________________ is equivalent to __________________

__________________ is equivalent to __________________

13.10.7 Wrap-Up: Reflection

1. Rate yourself using the scale below on today's learning target.

   I can verify using substitution that linear expressions are equivalent.



2. Write an explanation of why you rated yourself where you did.

Unit 13, Lesson 11: What Values Make an Equation or Inequality True?

Benchmark

MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.

Learning Target

- [ ] I can determine if a given integer value makes an equation or inequality true.

13.11.1 Warm-Up: Day at Work – Film Composer

Shohan Cagle is a film composer. He writes music for commercials and soundtracks for TV shows and movies.

1. What did you notice about what he does to write a soundtrack?

2. Shohan said that most people say they don't normally hear the soundtrack. Name a TV show or a movie whose soundtrack you remember.

3. Shohan mentioned that one of his challenges is a short deadline. He said that his strategy to get his work done is "to sit and focus and get in the right mindset". How can you use Shohan's strategy?

13.11.2 Guided Instruction: Determining Values That Make an Equation True or False

An ***equation*** is a mathematical relation statement where two equivalent expressions and values are separated by an equal sign.

1. In the equation $4x + 5 = 8x - 23$, what are the two expressions?

2. Work with your partner to guess a value for x and then substitute it into both expressions. A high and a low guess have been provided for you in the table.

x	$4x + 5$	$8x - 23$	Are the expressions equal?
$^{-}3$	$4(^{-}3) + 5 = {}^{-}7$	$8(^{-}3) - 23 = -47$	No
14	$4(14) + 5 = 61$	$8(14) - 23 = 89$	No

For an equation to be **true**, both sides of the equation must have the same value. For example, the equation $78 - 56 = 2 \cdot 11$ is true because both expressions, $78 - 56$ and $2 \cdot 11$, are equal to 22.

The ***solution*** of an equation is the value(s) that, when substituted in for the variable(s), makes the equation true.

3. Based on your guesses, what is the solution to the equation $4x + 5 = 8x - 23$?

4. Use substitution to determine if the equation $2x + 4x + 7 = 19$ is true for each value.

x	$2x + 4x + 7 = 19$	Is the equation true?
-1		
0		
2		

5. The solution to the equation $2x + 4x + 7 = 19$ is $x =$ _____ because when this value is substituted for x, the equation is _________.

13.11.3 Guided Practice: Determining Which Values Make an Equation True or False

From the specified set of integers, determine which, if any, make the equation true. Show your work.

1. $18 + 7w = 10w - 9 \quad \{-9,\ -2,\ 4,\ 9,\ 11\}$

2. $-2(3 - p) = 4p + 10 \quad \{-8, -3, 0, 10\}$

13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False

An ***inequality*** is a mathematical relationship between two expressions that are not equal ($<$, $>$, or $\neq$) or are not always equal ($\leq$ or $\geq$).

A ***strict inequality*** is a mathematical relationship between two expressions that are not equal. A strict inequality uses the symbols, $<$, $>$, or $\neq$.

1. Use substitution to evaluate the mathematical statement. Then determine if each inequality is true for the value of x.

x	**Strict Inequality** $9x - 5 < 13$	**Inequality** $9x - 5 \leq 13$	**Which inequality is true?**
5			
2			
$^{-}3$			

2. Answer the following.

   a. The inequality $x < 5$, means any number that is

      ○ smaller
      ○ larger

      than 5 will make the inequality true.

   b. List four numbers that are not integers that make the inequality true.

3. Answer the following.

   a. The inequality, $-11 \geq x$, means any number that is

      ○ smaller than or equal to
      ○ larger than or equal to

      $^{-}11$ will make the inequality true.

   b. List four numbers that are not integers that make the inequality true.

4. Use substitution to determine if the inequality $16 \geq 7m - 5$ is true for each value specified for m.

m	$16 \geq 7m - 5$	**Is the inequality true?**
-2	$16 \geq 7(-2) - 5$ $16 \geq -14 - 5$ $16 \geq -19$	Yes
0		
3		
6		

5. Answer the following.

a. Determine the integer(s) from the specified set that make the inequality **true**. Show your work.

$-x + 3 + 5x \geq {}^{-}8x - 7 \qquad \{-12, -5, 5, 16\}$

b. When Diego substituted 16 in $-x + 3 + 5x \geq {}^{-}8x - 7$, he stopped after writing $-16 + 3 + 5(16) \geq {}^{-}8(16) - 7$ because he said he knew it was true. When his teacher asked him how he knew, he said it was because the left side would be a positive number and the right side would be a negative number.

With your partner, discuss how Diego knew the left side of the inequality would be a positive number and the right side of the inequality would be a negative number.

6. Determine which values from the specified set make the inequality **false**. Show your work.

$12 < 2(13y - 1) \qquad \{-14, 0, 1, 9\}$

The values from the specified set that make the inequality $12 < 2(13y - 1)$, **false** are ____________________.

13.11.5 Your Turn!

1. Use substitution to determine if the inequality $86 \geq 17a - 8a - 23$ is true or false for each integer value specified for a.

a	$86 \geq 17a - 8a - 23$	True or False
$^{-}8$		
0		
7		
12		
21		

13.11.6 Wrap-Up: Reflection

1. One question I still have after today's lesson is …

____

____

____?

2. Today I am still unsure about …

____

____

____.

3. To fill in this gap I intend to …

____

____

____.

Unit 13, Lesson 12: Equivalent Expressions Using Properties of Operations

Benchmark

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

Learning Targets

- ☐ I can use properties of operations to write equivalent forms of an algebraic expression.
- ☐ I can determine whether two linear expressions are equivalent.

13.12.1 Warm-Up: Fact or Fiction?

1. Determine whether each equation below is fact (true) or fiction (false).

Equation	Fact or Fiction
$(30+8)\cdot 9 = 270 + 72$	○ Fact ○ Fiction
$4 + 5(9-15) = 9(9-15)$	○ Fact ○ Fiction
$2\cdot(7+8) = (2\cdot 7)+8$	○ Fact ○ Fiction
$(5+4)-10 = 5+(4-10)$	○ Fact ○ Fiction

13.12.2 Refresher: Rewriting Expressions Using Properties of Operations

1. Fill in the missing blanks for each property.

   a. Using the ***commutative property of addition***, $3 + 12$ can be rewritten as ________________. Both expressions have a sum of ______________.

   b. Using the ***commutative property of addition***, $4x + 5$ can be rewritten as ______________. Both expressions will have the same sum regardless of the value of x.

   c. Using the ***commutative property of multiplication***, $(4)(-2)$ can be rewritten as _____________. Both expressions have a product of ____________.

   d. Using the ***commutative property of multiplication***, $(-3y)(0.5)$ can be rewritten as _______________. Both expressions will have the same product regardless of the value of y.

   e. Using the ***associative property of addition***, $-5 + (6 + 1)$ can be rewritten as _____________. Both sums are equal to ______________.

   f. Using the ***associative property of addition***, $9x + (-2x + 5)$ can be rewritten as _____________. Both expressions will have the same sum regardless of the value of x.

   g. Using the ***associative property of multiplication***, $2 \cdot (7 \cdot 3)$ can be rewritten as _____________. Both products are equal to _____________.

   h. Using the ***associative property of multiplication***, $(-6)(4y)$ can be rewritten as _____________. Both expressions will have the same product regardless of the value of y.

   i. Using the ***distributive property of multiplication over addition***, $5(6 + 2)$ can be rewritten as _____________. Both expressions have a value of ______________.

   j. Using the ***distributive property of multiplication over addition***, $7(3y + 8)$ can be rewritten as _____________. Both expressions will have the same product regardless of the value of y.

2. Discuss with your partner why the associative property and the commutative property do not hold true for subtraction and division. Give an example of each to show mathematically that these properties do not apply to subtraction and division.

3. Jay gave the example $6 + {}^{-}11$ is equal to ${}^{-}11 + 6$ to show that subtraction is commutative. Explain to Jay why his example does not show that subtraction is commutative.

13.12.3 Guided Instruction: Equivalent Expressions Using Properties of Operations

Algebraic expressions are considered equivalent if when evaluating the expression by substituting values in for variables, the resulting values are the same.

Algebraic expressions are also considered equivalent when one expression is rewritten using one or more of the properties of operations.

1. Complete each statement.

   a. The
      - ○ associative
      - ○ commutative
      - ○ distributive

      property can be used to state that $5(x+2) - 3x$ is equivalent to $5x + 10 - 3x$.

   b. The
      - ○ associative
      - ○ commutative
      - ○ distributive

      property can be used to state that $\frac{2}{3}x + \left(\frac{3}{8} + \frac{5}{6}x\right)$ is equivalent to $\frac{2}{3}x + \left(\frac{5}{6}x + \frac{3}{8}\right)$.

2. Which properties can be used to show that $2(x-4) + 5(3+6x)$ is equivalent to $2x + 30x - 8 + 15$?

13.12.4 Guided Practice: Rewriting Expressions Using Properties of Operations

1. Write an equivalent expression using a property of operations. State the property used.

Expression	Equivalent Expression	Property Used
$-6(2-3x)+1-\frac{x}{2}$		
$12x+10+4x-17$		
$5x+(3x+9)+2$		

2. Several expressions are given below. Determine which expressions are equivalent using properties of operations.

$35x+(7x+56)$	$5(8x+7)$	$35x+56$
$(8x+7)\cdot 5$	$7(5x+8)$	$(35x+7x)+56$

Complete the table by writing the expressions that are equivalent based on the property used.

Equivalent Expressions		Property Used
		Associative Property
		Commutative Property
		Distributive Property

13.12.5 Your Turn!

Write an equivalent expression using the associative, commutative, or distributive property. State the property used.

1. $10x+\frac{2}{5}(-30+15x)$

2. $(18x+15)+4x-17$

3. $\left(-\frac{1}{5}x+6\right)\cdot 5$

4. $\frac{5}{4}n + \frac{7}{3} - \frac{11}{2}n - 12\frac{1}{3}$

5. $-3\left(\frac{1}{2}x + 4\right) - \frac{3}{5}(-1 + 10x)$

13.12.6 Wrap-Up: Ticket Out the Door

Use properties of operations to determine if the expressions are equivalent.

1. $\frac{4}{5}(x - 1) - \frac{1}{8}(2x + 1)$ and $\frac{11}{20}x - \frac{37}{40}$

2. $1\frac{3}{10}x + \frac{1}{5}x - \frac{1}{10} - x - \frac{1}{10}$ and $1\frac{1}{10} - \frac{4}{5}x$